High-Dimensional Computing for aDBS

Brain Hacks 2026

Exploratory analysis + Experimental Project & Benchmarking

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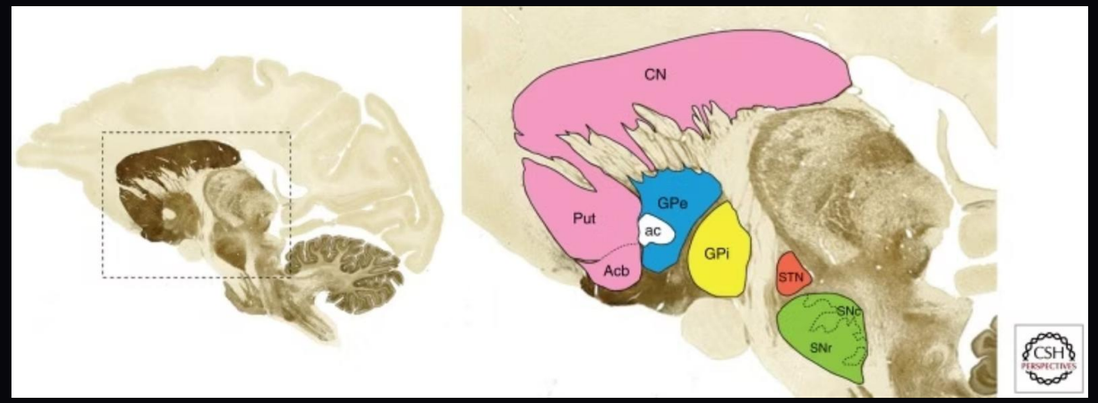
STN & GPe Interactions

- Synaptically coupled subthalamic nucleus (STN) & globus pallidus externa (GPe) cells
- STN excite
- GPe inhibit
- Normally both fire independently, synchronized firing is a beta band (13-30 Hz)
- Synchronized activity is detectable as signal in the local field potential (LFP) in the Basal Ganglia



José L Lanciego, Natasha Luquin, José A Obeso, Cold Spring Harb Perspect Med. 2012

ODE Simulation

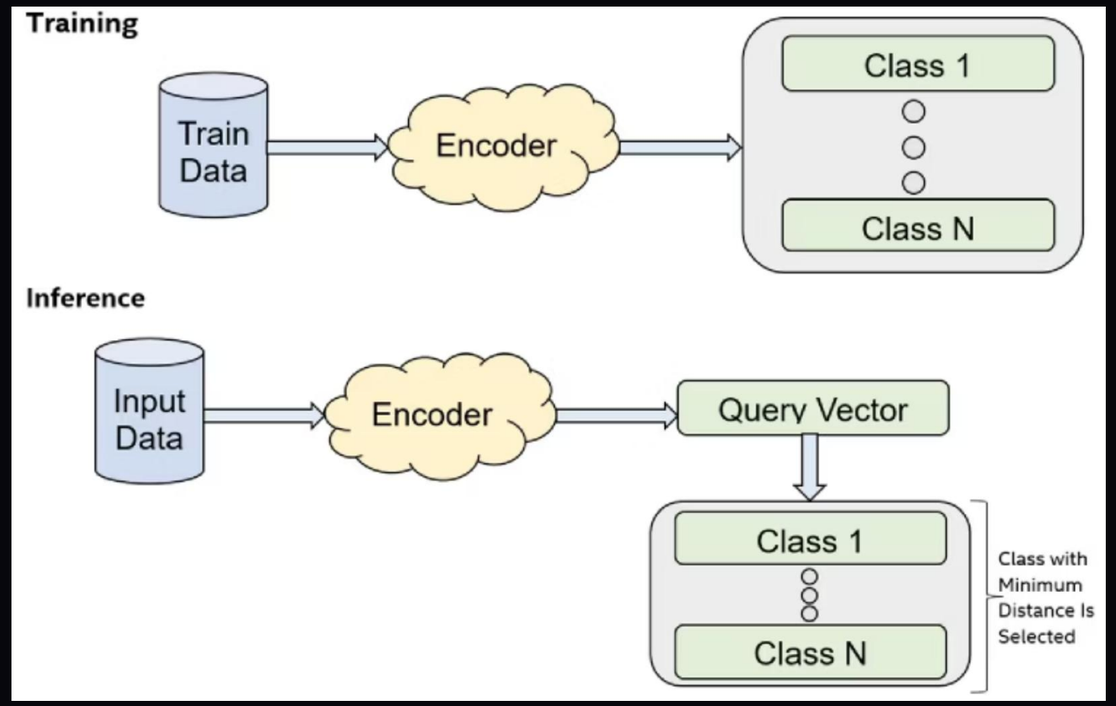
- Set of ordinary differential equations modeling how this system changes over time
- Set biological rules that dictate simulated neuron changes of voltage and ion channels
- Use gating variable (0-1) to represent "openness" of a synapse

Importance

- Deep brain stimulation of STN cells can be used as a therapy for Parkinson's disease.
- Usually classical threshold controller, we propose using an adaptive approach using hyper vectors instead.
  - With the benefit of not continuously running stimulus, potentially lower side effects
- Proof of concept run on the synthetic data from ODE simulation

What is HDC?

- One-Shot Learning: single data pass
- Encodes data into high-dimensional representations (large vector of 0s and 1s called a hypervector)
- Inference: encodes incoming data and computes cosine similarity
- All bitwise operations (flipping 0s and 1s) can be done billions of times per second by a CPU, which makes hypervectors highly efficient and attractive for edge ML



Objective

> Can HDC match a classical beta-power controller for adaptive DBS while being cheaper to run?

- Build a biophysical ODE model of the STN-GPe loop that produces healthy and Parkinsonian dynamics
- Encode LFP signal windows into hypervectors to detect pathological brain state in real time
- Compare HDC controller against classical beta-threshold aDBS, continuous DBS, and no stimulation
- Measure control quality (beta suppression, pathological occupancy) and efficiency (duty cycle, decision latency)

System Design

STN-GPe ODE Model + Data Pipeline

- 8 STN + 8 GPe neurons, Hodgkin-Huxley conductance-based equations
- Increasing GPe-STN inhibitory coupling drives the network into pathological beta synchrony (13-30 Hz)
- Population-mean STN membrane voltage used as LFP surrogate

"Training": Hypervector Encoder Search

- Grid search over 24 configurations: dimensionality, quantization bins, value hypervector initialization
- Each LFP window is normalized, quantized, bound with position hypervectors, and bundled into one hypervector
- Best encoder selected by balanced accuracy on healthy vs pathological windows

Closed Loop Controller

- Every 50ms, compute similarity margin between query hypervector and healthy/pathological prototypes
- Dual-threshold hysteresis: stimulation turns on above one threshold, only turns off below a lower one
- Compared directly against a classical beta-power threshold controller using the same on/off logic

Our Process

Sample Data + Validate ODE model

- Run healthy and pathological simulations
- Confirm beta power is clearly elevated in pathological LFP before proceeding
- Critical checkpoint: if the two regimes are not separable here, nothing downstream works

"Train" Models

- Encode windowed LFP segments into hypervectors using best encoder config
- Bundle healthy windows into one prototype, pathological windows into another
- Single pass through training data, no iterative optimization

Offline Eval

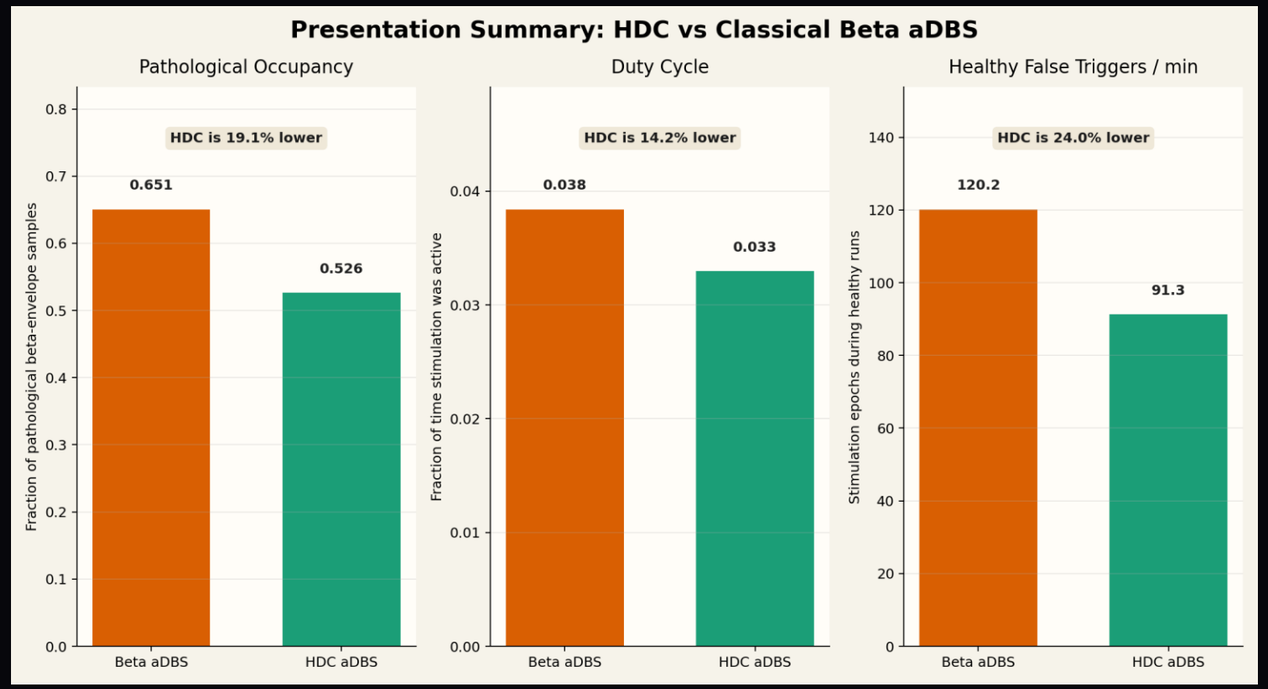
- Test cosine similarity margin separability across healthy, pathological, onset, and recovery windows
- Onset and recovery windows stress test the encoder on ambiguous transitional states

Online Eval (Closed Loop Controller)

- Run all 4 conditions on the same pathological onset seed
- Compute beta suppression, pathological occupancy, duty cycle, and per-window decision latency

Findings

- Generated transition subsets (onset, recovery, moderate coupling) to simulate more realistic data beyond clean healthy and pathological extremes
- Grid searched 24 encoder configurations to find optimal hypervector encoder hyperparameters
- Compared logistic regression over raw hypervectors vs prototype hypervectors built from bundled encodings
- Prototype approach achieved ~60% balanced accuracy with ~0.5-0.7ms average inference latency
- Accuracy gap likely due to the encoder not fitting the LFP data structure optimally rather than HDC itself failing
- HDC pathological occupancy, duty cycle, healthy false triggers/min are 19.1%, 14.2% and 24.0% lower respectively



Next Steps

- Try bootstrapping and fitting weak learners as hypervectors instead of raw window encodings
- Replace manual grid search with SGD-based encoder optimization for higher classification accuracy
- Alternatively train a linear classifier directly over hypervectors to better fit the data while keeping inference speed equivalent
- Run full 4-condition closed-loop comparison and complete online evaluation metrics

References

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Code Repository:

- https://github.com/Thomson-Lam/HDC-aDBS

Thank you for listening!

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